



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering
Academic Year: 2019- 2020 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. Computer Science and Engineering.
Course Code & Title: CS8601 Mobile Computing
Name of the Faculty member: Dr.M.Kaliappan, Associate Professor/CSE

Theme of discussion: GSM Services and Architecture.

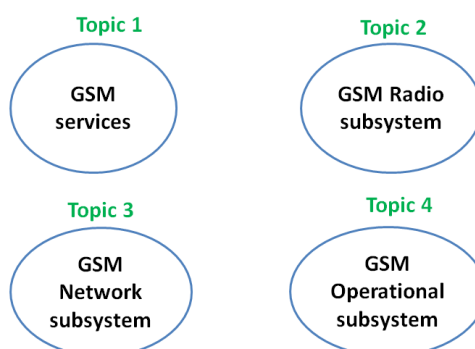
Date and Time: 10.02.2020 & 4.10 pm to 5.00

2 Innovative practices: JIGSAW Collaborative Learning

“Jigsaw”[1][2] is a collaborative group activity with a twist: students effectively teach each other (with the teacher’s guidance). Students learn through the process of communicating with one another about a given skill or procedure, topic or problem.

2.1 Topic: GSM Services and Architecture

Sub topics



2.2 Outcome

- Describe the services and functionalities of GSM architecture.

2.2 Justification for choosing the topic

This topic describes the core architecture of mobile communication. These core principles are used as bases for designing 3G and 4G systems. This topic and its related components are more suitable for a Jig-Saw cooperative learning activity in which students get exposure in all GSM services and components. Students can master all the core components of GSM. Students should be able to enhance their communication, peer learning, individual and teamwork, and self-learning by this activity.

2.3 Procedure and Implementation

2.3.1 Team formation

The heterogeneous groups with six members were formed. Each group formed by bright students, average students, and slow learners, consists of 4 members. One group contain 5 members. Classification of students such as *a bright student, average students, and slow learners* is based on their academic performance. The total class strength is 53 (Boys

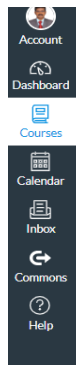
and Girls). **Thirteen heterogeneous groups** were formed based on academic performances such as the CGPA of end semester examination. The table1 shows the name of the groups.

Sno	Group name	No. of members
1	Killer boys	4
2	Evil boys	4
3	Blooms	4
4	Five stars	5
5	Fantastic Four	4
6	Mars	4
7	Lightning crackers	4
8	Nameless	4
9	Phoenix	4
10	Power puff girls	4
11	Team 1	4
12	Team 3	4
13	Team 4	4

The figure 1 and Figure 2 shows the group name and its team leader which is created in Canvas.

The screenshot displays the Canvas LMS interface. On the left, a sidebar contains navigation links: Grades, People, Pages, Files, Syllabus, Outcomes, Quizzes, Modules, Conferences, Collaborations, Attendance, New Analytics, and Settings. The main content area is divided into two sections: 'Unassigned Students (1)' and 'Groups (13)'. The 'Unassigned Students' section features a search bar and a list of users, including M.Kaliappan. The 'Groups' section lists 13 groups with their names, team leaders, and student counts. The groups are: 1. Killer boys (4 students, leader SUGUN M), 2. Evil boys (4 students, leader A.J.VAISHNAVIDEVI), 3. Blooms (4 students, leader MOHAMEDRISHWAN J), 4. Fantastic Four (4 students, leader YESHWANTHINI S), 5. Five star (5 students, leader SIVA.RANJINI R), 6. Killer boys (4 students, leader GOWTHAM P), 7. Lightning Crackers (4 students, leader ANUSHREE M), and 8. Mars (4 students, leader SELVA ISHWARYA R). The interface also shows a 'Full' status for each group and a 'Go to Settings' link at the bottom right.

Figure 1: group name and its team leader



4 / 4 students		
▶ Nameless	RAJARAJESWARI P	Full 4 / 4 students
▶ Phoenix	dharsini1242000@gmail.com	Full 4 / 4 students
▶ Power Puff Girls	SASI REKHA M	Full 4 / 4 students
▶ Team 1	NARMADHA S	Full 4 / 4 students
▶ Team4	Prabhu M	Full 4 / 4 students

Figure 2: Group name and its team leader

Step 1

- Thirteen groups with four/five members were formed based on the students' academic performance and ability. It is called home group.
- One student from each group selected as the leader. Each student in a group was given a number from 1 to 4. Each number is representing the following four sub topics. A student was assigned one subtopic in a home group.
 1. GSM services
 2. GSM Radio system
 3. GSM Network sub-system
 4. GSM operational sub-system

Step 2

- Expert groups were formed that consist of student from home groups who were assigned the same number.
- Expert group members were given 15 minutes to discuss an assigned sub topic.

Step 3

- The expert group students brought back into their home groups and allotted 10 minutes for further discussion in which each member become expert in the entire GSM services and its components.

Step 4

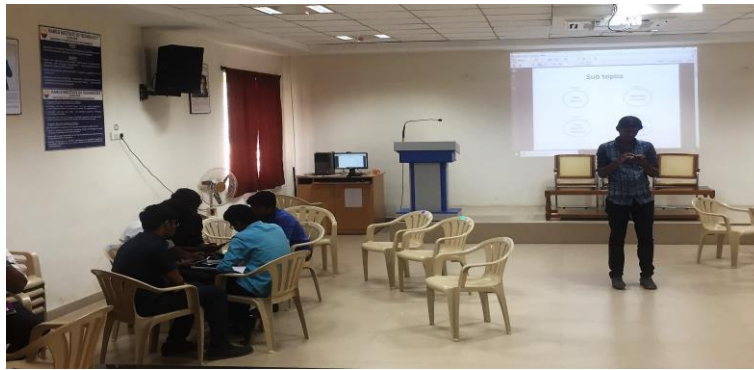
- 10 minutes were allotted for the presentation. One student from each home group presented his learned content.
- The others in the group were encouraged to raise queries for clarification

Step 5

- At the end of the session, feedback was collected in five minutes.

2.4 Glimpse





2.5 Reflective Report

2.5.1 Challenges

- Few boys are hesitated to form the expert groups with girls.
- Participation of slow learner for discussion on sub-topic.

2.5.2 Pre-Implementation

The procedure of this activity and team formation was announced in Canvas well in advance. Figure 3 show the announcement of Jigsaw activity in Canvas.

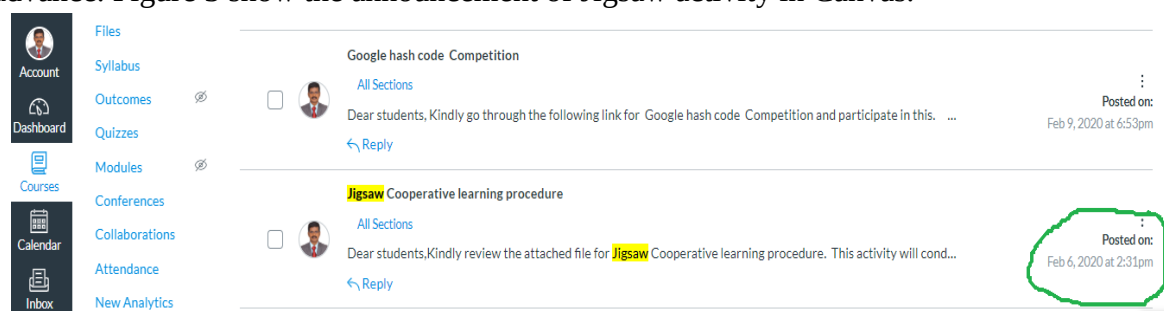


Figure 3 : Announcement of Jigsaw activity

2.5.3 Observations:

The students were provided the opportunity to become experts in particular sub topic. They shared the knowledge they gained with peers. This learning *promotes both self-learning and peers teaching* among the students to master the course content at a deeper level. Most of the students participated actively. The Jigsaw technique helped the students to develop expertise in a particular topic.

I motivated the students to form heterogeneous groups (boys and girls) based on gender. Also, I encourage the slow learners to refer the course materials during the discussion for active participation. One student from evil boys and Mars *expressed their learned knowledge very well*. The students were encouraged to raise queries for clarification during the presentation.

2.5.4 Students Response:

- Most of the students discussed with teammates comfortably during the discussion in expert groups.
- The bright students were actively involved with peers to discuss the topics in their home group.
- The slow learners wanted to assist during the discussion. They were allowed to refer the material about the topic of the discussion.

- Most of the students enjoyed the discussion in the home and expert group.
 - 90% of the students were given excellent feedback regarding the feedback session.
- Figure 4 shows the sample collected feedback.

2.5.5 Post Implementation

1. Based on the observation, a lot of *incidental learning* happening in the discussion forum. They observed and learned components of GSM architecture individually that lead to self-learning. They *actively involved* in this expert group discussion rather than it being a one-way communication.
2. All the home groups performed well in the group presentation session. This session *re-enforced their understanding*.
3. Slow learners from killer boys and Team4 group were not comfortable in their presentation due to lack of attention and participation. However, they understood the concepts of these learning methods.

Students' response ensures that the students could improve listening, and communication, skills

2.6 Relevance of program outcomes

Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Outcome	3							2	2	2		2

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Feed Back
Innovative practice: Jigsaw cooperative learning technique.
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Name of the Faculty member: Dr.M.Kaliappan, Associate Professor/CSE
Theme of discussion: GSM SERVICES AND ARCHITECTURE.
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Is encourage cooperative learning among yourself ? Yes No

Is Connection with course material in a creative and engaging way? Excellent Good

Is improve listening, communication, and problem-solving skills? Excellent Good

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Figure 4: Sample Feedback

Reference:

1. https://www.newcastle.edu.au/_data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf
2. <https://www.jigsaw.org/>
3. Mr.K.Vignesh Saravanan, OIT551- Database Management Systems, Topic: Modelling of ER diagram , Jigsaw activity, 11.07.2019

Faculty In charge

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